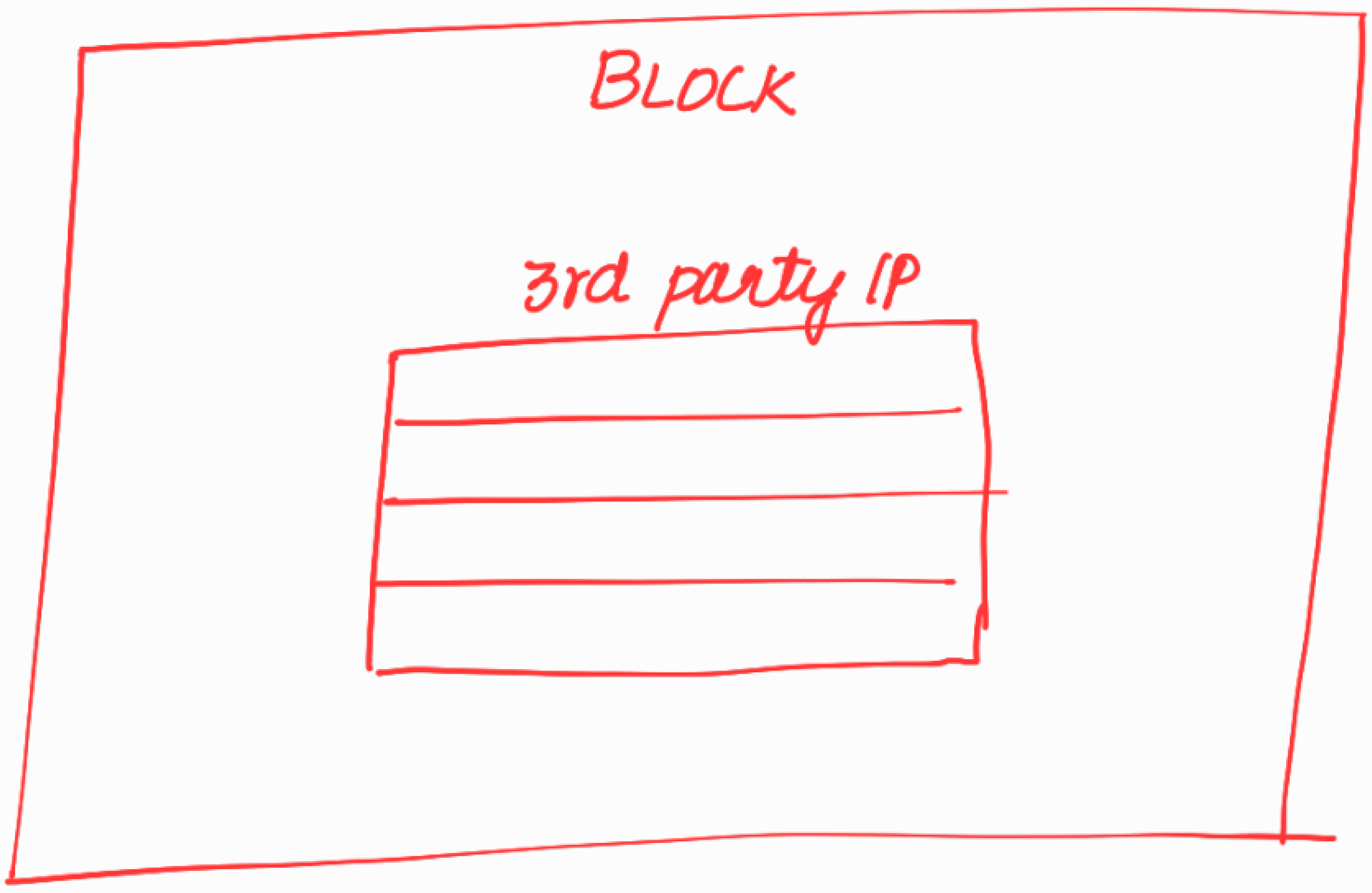
CASE 10:-

In a block, there can be some modules for which the scan chains are already stitched.

For eg:- 3rd party IP like USB.

↓

it will be bought from some vendors and it will be directly used in the design.



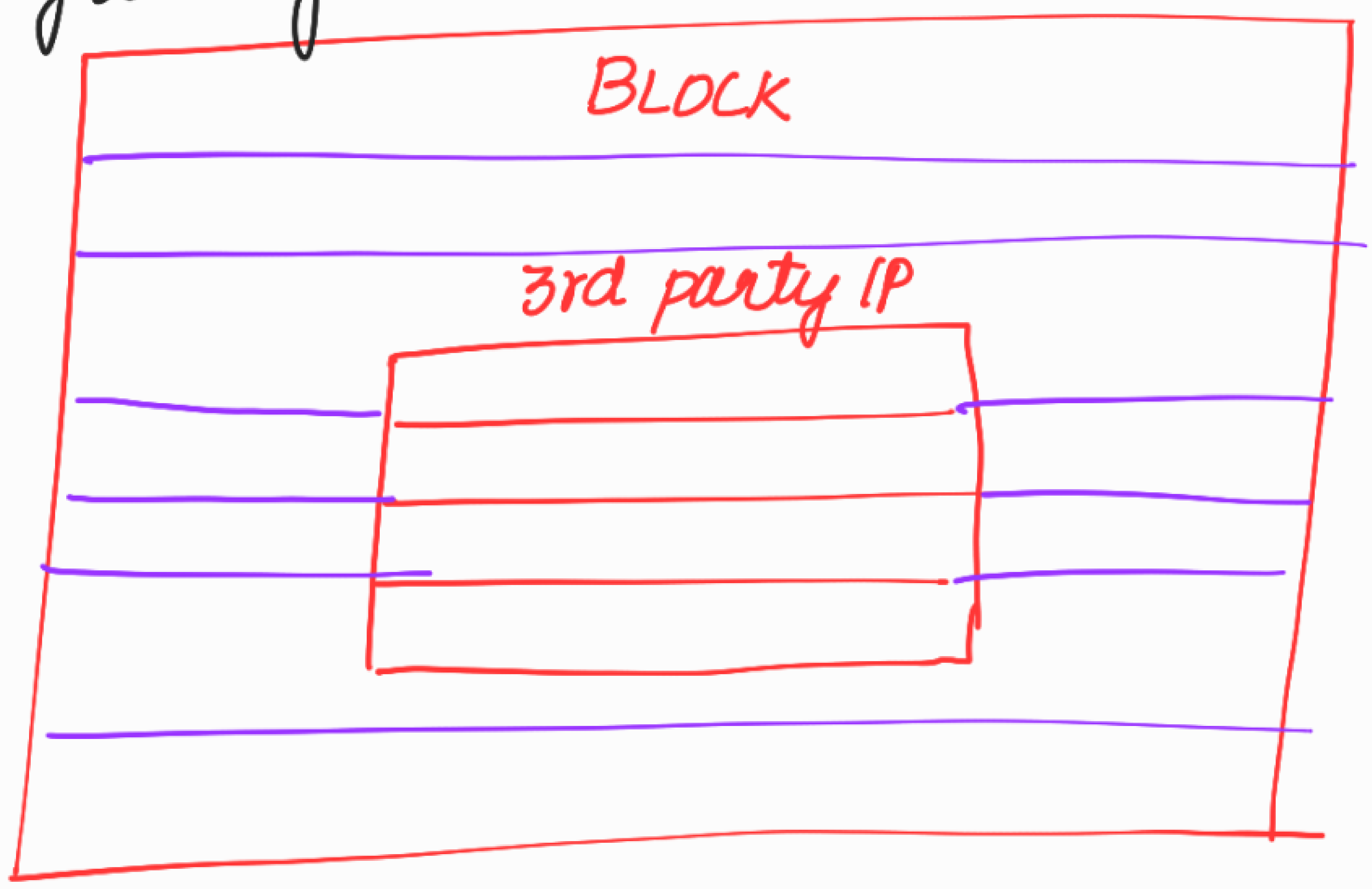
Pre-existing/Pre-defined scan chains (S6 violation)

We need to give info to the tool about the already existing scan chains

→ which module
→ sub-chain name
→ scan i/p, scan o/p port
→ no. of ffs in the scan chain.
→ Type of scan cell.
→ scan enable

→ name of the clock connected to the ffs in the pre-existing scan chains, offstate of the clock.

After giving the above info

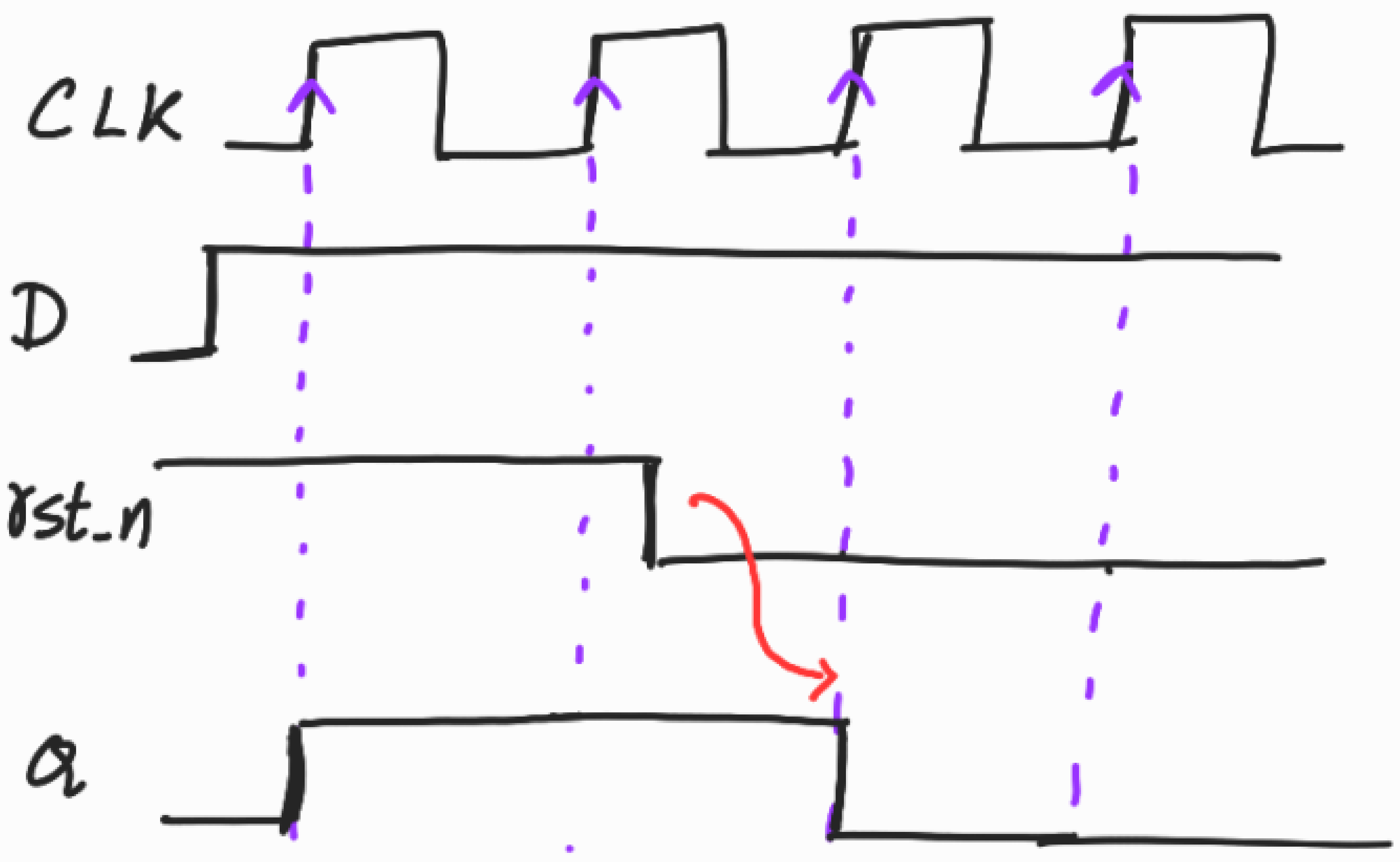


NOTE :-

2 types of FFS :-

(i) SYNCHRONOUS RESET FLOP

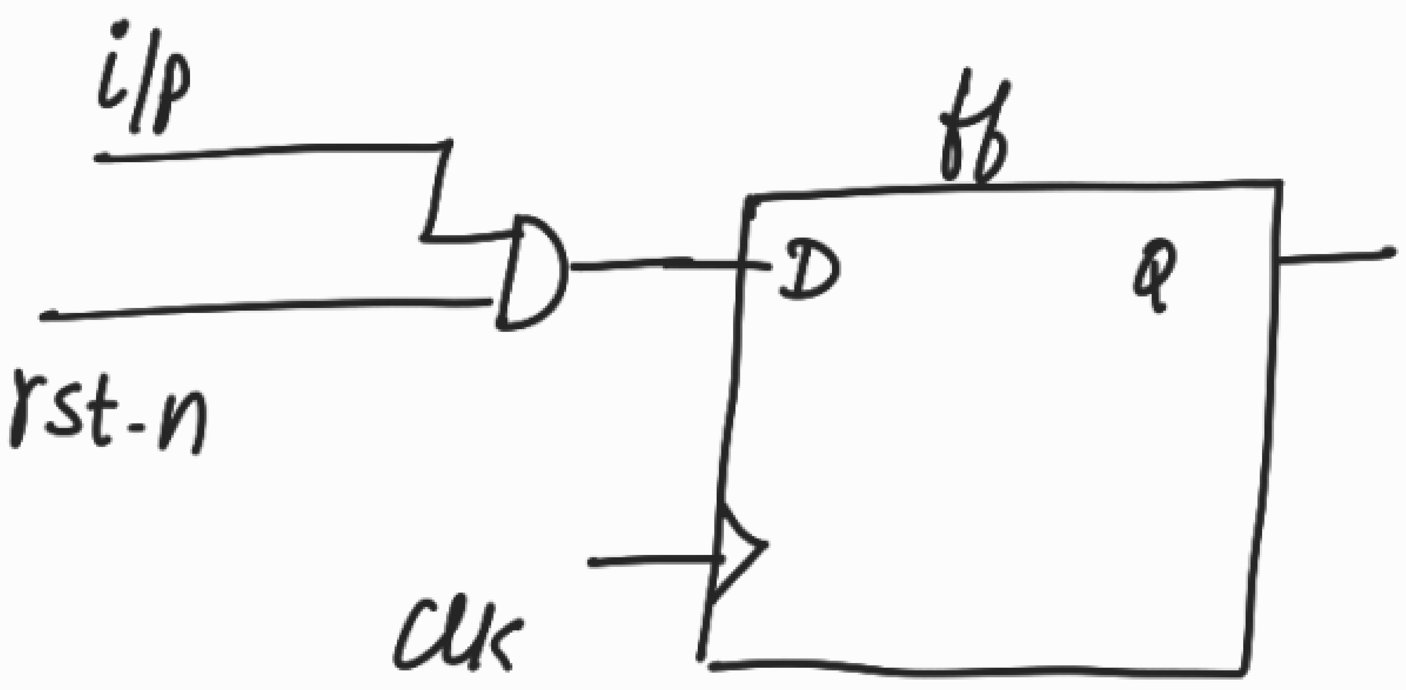
- Reset can be applied anywhere but its effect is only at the clock's active edge.



active low reset

$R=0 \Rightarrow Q=0$

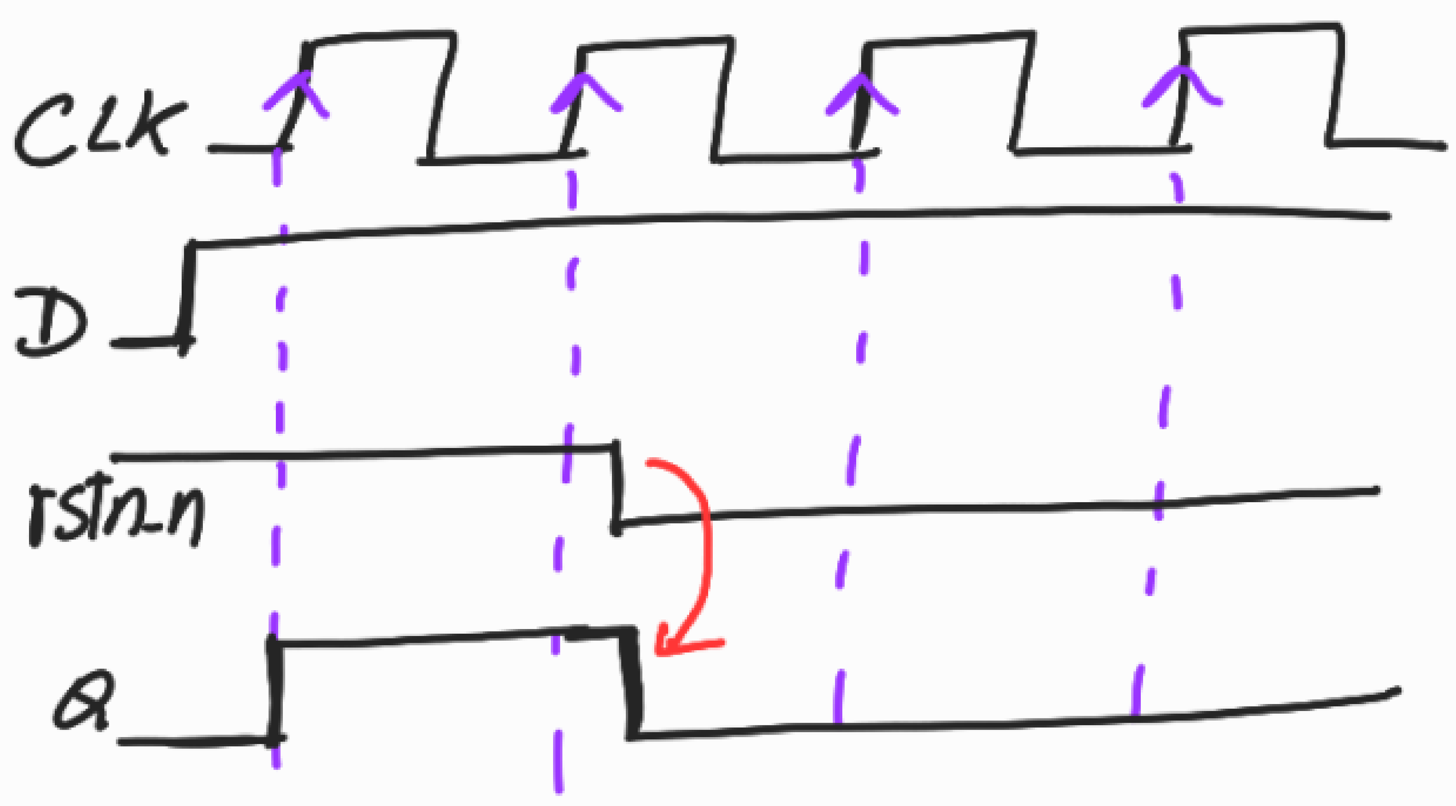
$R=1 \Rightarrow Q=D$



(ii) ASYNCHRONOUS RESET FLOP

- Reset can be applied anywhere and its effect is immediate.

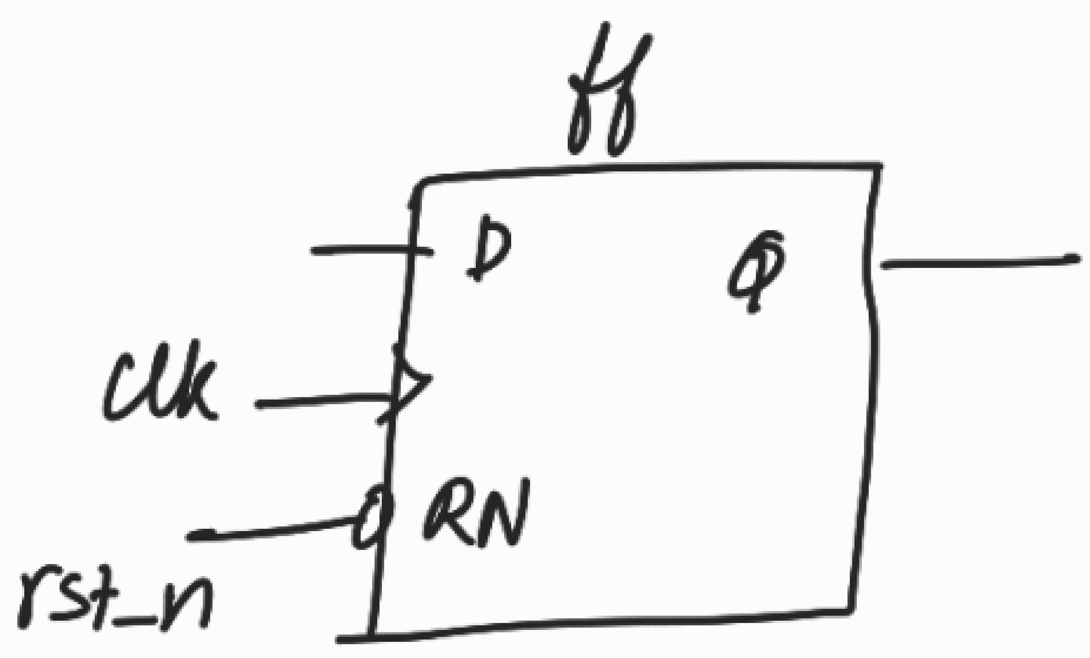
It is not synchronized with the clock.



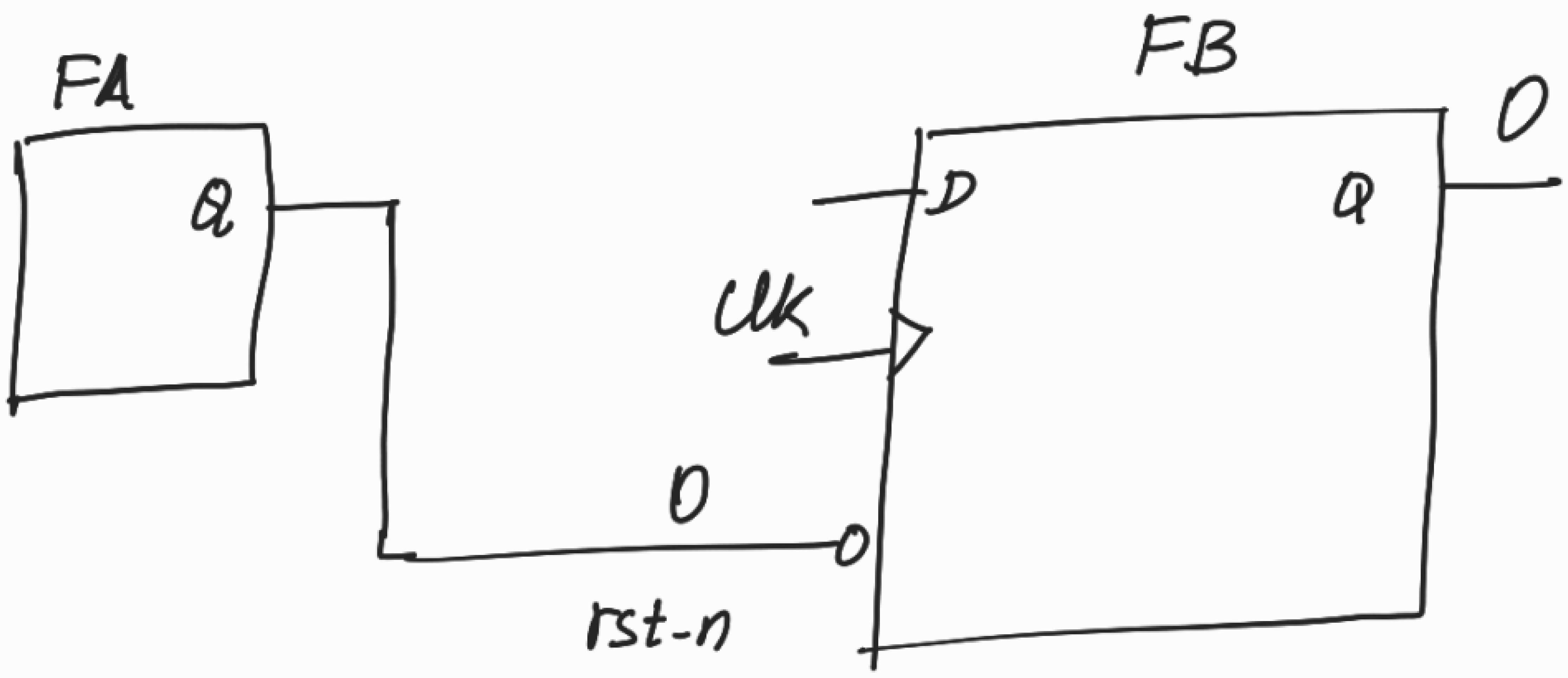
active low reset

$R=0 \Rightarrow Q=0$

$R=1 \Rightarrow Q=D$

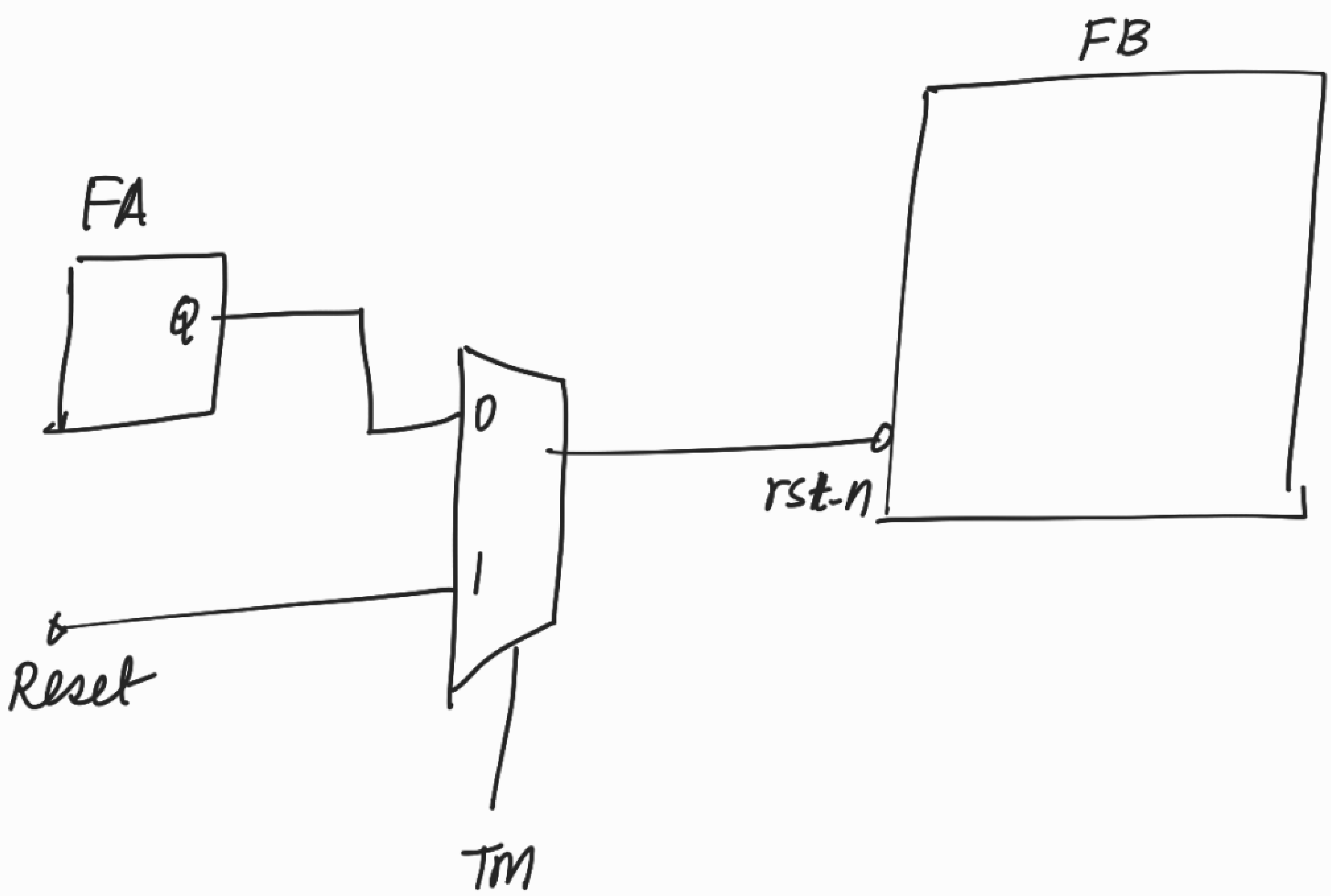


ASYNCHRONOUS RESET FLOP :-

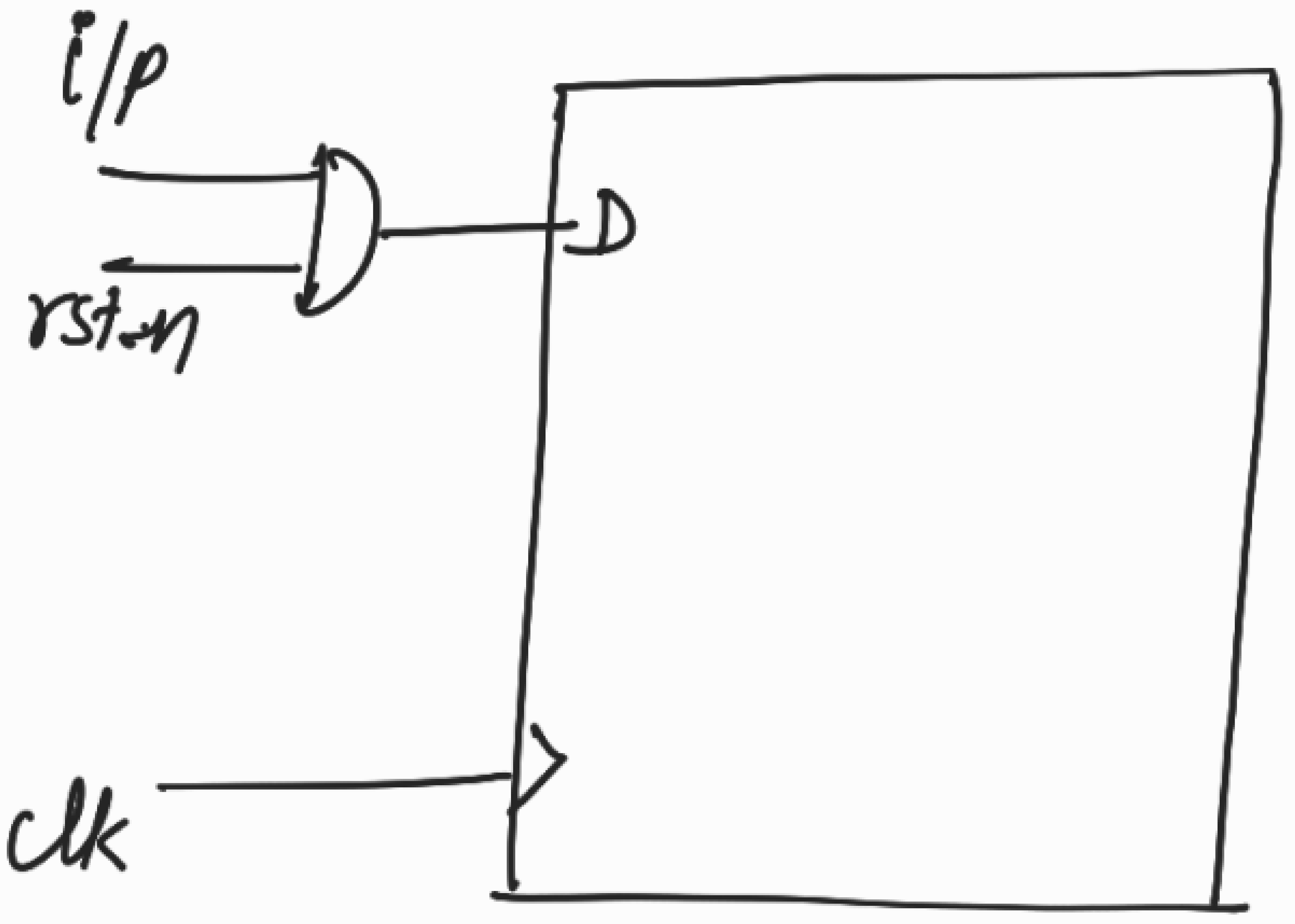


$\Rightarrow$ SI violation.

The o/p of FA can become 0.
So FB's rst_n can become 0.
$\therefore$ FB's $Q = 0$.
$\therefore$ Proper patterns will not be shifted into the scan chains.



SYNCHRONOUS RESET FLOP:-

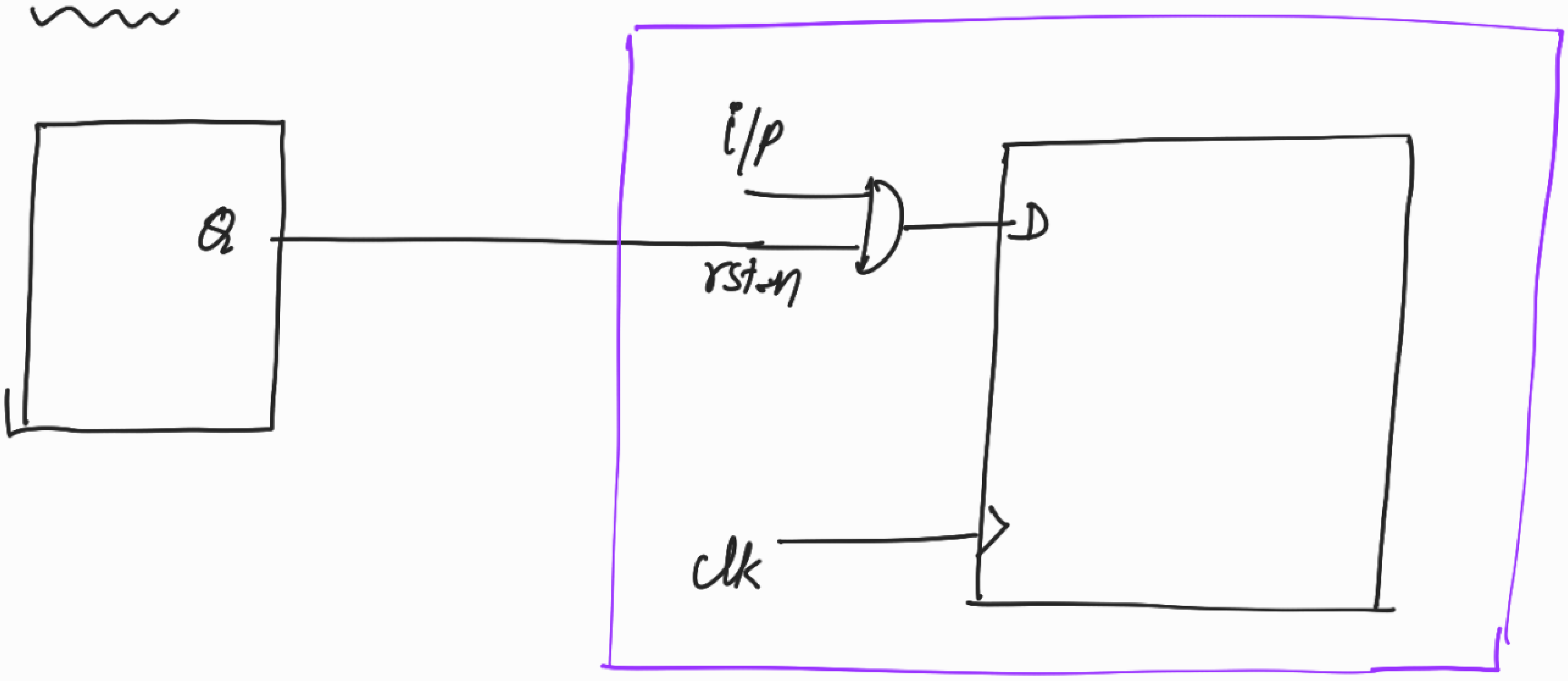


For sync reset flop, there is no separate pin for reset.

(i/p and rst_n) ⇒ D i/p of the ff.

So it will not affect the state of the flop during shifting.

CASE 10:



↳ Tool does not throw SI violation